Evaluator of **N**earby dwarf stars in high-**Z** photometrically selected **O**bjects (ENZO)

Contents

1 Introduction

`ENZO` is a contamination calculator designed to estimate the number of ultracool dwarfs (UCDs) that can mimic high-redshift galaxy candidates in specific observational fields.

The current version includes the following features:

- Two evolutionary models are available: *Sonora Bobcat* and *Sonora Elf Owl*.
- Photometric redshift fitting is performed using `EAzY`, with the `Hainline`, `V1p0`, `V1p3`, and `Larson` galaxy template sets. For stellar χ^2 fitting, the `SPEX` template library is adopted.
- Near-infrared (NIR) color-selection criteria are available from Roberts-Borsani et al. (2022) and Bouwens et al. (2015).
- For the spatial number density distribution of UCDs in the Milky Way, the model allows the adoption of the density distribution from Aganze et al. (2022).

This document serves as a guide for setting up and using the *ENZO* software. Detailed descriptions of the analysis methods and theoretical background are presented in **?**.

2 Installation

After downloading the project folder, you will find the following files and directories:

- `data_set`
  This directory contains all datasets required by the program:
  - `1Mj_fit.dat`
    This file contains the effective temperature values corresponding to the space density, ρ_0, derived from the fit presented in **?** (Figure 29).
  - `Bobcat_model`
    This directory contains the photometric and spectral models adopted from: [1].
  - `Elf_Owl_photometry`
    This directory contains fluxes (with a zero point of 25) and AB magnitudes for metallicities of -1.0, -0.5, $+0.0$, $+0.5$, $+0.7$, and $+1.0$. Each file corresponds to a carbon-to-oxygen ratio (C/O) of 1.0.
  - `transmission_filters`
    This directory contains the transmission profiles of all filters used in the program. The filter profiles were obtained from the SVO Filter Profile Service: [2].
- `Results`
  This directory is initially empty. All outputs generated by the program will be saved here.

[1] https://zenodo.org/records/5063476
[2] https://svo2.cab.inta-csic.es/theory/fps/

- `example_param.yaml`
  An example parameter file that users can modify to create their own input configuration.

- `Full_program.py`
  This is the main executable script. Running this file will launch the graphical user interface (GUI) for the program.

- `FULL_synthesis_def.py`
  This file contains all function definitions required by the program. In particular, users may need to modify the `run_eazy()` function to update the template-file path and the number of prior filters used.

- `param_GUI.py`
  This file contains the code responsible for the parameter-input interface and other GUI widgets.

3 Parameter and Input file

There are two ways to create the parameter file. The first method is to generate it by running `FULL_PROGRAM.py` in your local environment. More details about the automatic setup procedure can be found in Section 4. The second method is to use the `example_param.yaml` file as a template for creating and configuring the simulation parameters.

3.1 Initial Setup and Classification Criteria

This section is divided into two parts: the initial setup and the classification criteria configuration. The initial setup requires the following parameters:

- `[fields_name]`: Specifies the target field name. This parameter may also be used in the output path. Provides a more detailed field identifier, such as `0959+0200` or `0341-6712`.

- `[RA]`: Right ascension of the target field in degrees.

- `[Dec]`: Declination of the target field in degrees.

- `[solid_angle]`: Converts an area specified as `(value, unit)` into steradians. For example, `[9.7, arcmin2]`. Supported units include:

  ```
  'sr', 'steradian', 'steradians', 'deg2', 'deg^2', 'square_degree',
  'square_degrees', 'deg', 'arcmin2', 'arcmin^2', 'arcminsq', 'arcminsquared',
  'arcsec2', 'arcsec^2', 'arcsecsq', 'arcsecsquared'
  ```

- `[filters_program]`: Specifies the filter names located in the `transmission_filters` folder. Filters for *JWST*, *HST*, and *Roman* are currently supported.

- `[mh_list]`: A list of metallicity values corresponding to the stellar model grid. Available values are:

  `'+0.0', '-0.5', '-1.0', '+0.5', '+0.7', '+1.0'`

- `[stellar_model]`: Specifies the stellar evolutionary model to use. Available options are `'Elf_Owl'` and `'Bobcat'`.

- `[ENZO path]`
  Specifies the location of the ENZO installation directory. This path should correspond to the root folder of the program and should end with `ENZO`. For example:

  `/Users/.../ENZO`

- `[Magnitude shift settings]`: Defines the magnitude range and bin size used to construct the detection-band grid. This setting requires three parameters:

  `mag_detectionband_start`,
  `mag_detectionband_final`,
  `mag_detectionband_bin`

3.2 Image Calibration Parameters

This section outlines the key parameters derived from image calibration for each band within a specific observational field, including:

- `[bands]`: A list of filter names.
- `[PHOTPLAM]`: Pivot wavelength in Å.
- `[PHOTFLAM]`: Photon flux density in units of erg s^{-1} cm^{-2} $Å^{-1}$.
- `[EXPTIME]`: Exposure time in seconds.
- `[depths]`: A dictionary containing the limiting magnitudes corresponding to each filter band.

These parameters vary between observational fields and are essential for generating the theoretical Poisson noise model (see the discussion in the "Noise Perturbation of Flux at the New Magnitude" section of ?).

3.3 Selection Method Settings

This section defines the parameters required for the selection methods. The general control parameters are:

- `[redo_mockcatalog]`: A Boolean parameter (`True` or `False`) that specifies whether the mock catalog generation should be rerun.

- `[redo_eazy]`: A Boolean parameter (`True` or `False`) that determines whether the `EAzY` photometric redshift fitting process should be rerun.

- For color-selection methods, no additional `redo` control parameter is required. The selection only requires the name of the adopted color-selection criteria.

Each selection method also requires additional method-specific parameters.

3.3.1 Photo-z Using `EAzY`

This method requires the `[templates]` parameter, which specifies the list of templates used for spectral energy distribution (SED) fitting. The template list should include both galaxy and stellar templates.

Available galaxy templates include:

`'Hainline'`, `'Larson'`, `'V1p3'`, and `'V1p0'`

For stellar fitting, the available template currently includes:

`'spex'`

3.3.2 Color–Color Selection

This method uses predefined color-selection criteria for source classification. If not specified manually, the default options include the selection criteria from Roberts-Borsani et al. (2022) and Bouwens et al. (2015).

Users may also define custom classification criteria. For the Bouwens et al. (2015) criteria, three predefined configurations are available, each optimized for different survey fields:

- `set_a`: Designed for *XDF*, *HUDF09-Ps*, and *CANDELS-GS+GN*.
- `set_b`: Designed for *ERS* and *BoRG/HIPPIES*.
- `set_c`: Designed for *CANDELS-UDS*, *COSMOS*, and *EGS*.

For the Roberts-Borsani et al. (2022) criteria, only `set_b` is available.

4 Automatic Program Setup

The automatic setup mode is designed for users who prefer to configure and run the program through a graphical interface rather than manually editing a parameter file. In this mode, the program guides the user through the configuration process and can automatically generate the required `.yaml` parameter file.

To launch the program, execute:

`FULL_PROGRAM.py`

After startup, the program will ask whether an existing parameter file is available.

4.1 Using an Existing Parameter File

If a parameter file has already been created, select `Yes`. A file browser window will appear, allowing the user to select the desired `.yaml` file. Once selected, the configuration contained in the file will be loaded automatically.

The user may then:

- Run the synthesis directly using the loaded configuration.
- Review and modify any parameters before execution.
- Save the updated configuration as a new `.yaml` file if desired.

4.2 Creating a New Parameter File

If no parameter file is available, select `No`. The program will open a configuration interface and guide the user through the required input parameters.

The user will be asked to provide information such as:

- Installation and data paths.
- Survey and filter selections.
- Model parameters.
- Simulation settings.
- Output options.

After all required information has been entered, the program will generate a new `.yaml` parameter file that can be reused in future runs.

4.3 Simulation Control Window

After a valid parameter file has been loaded or created, the simulation control window will appear.

This window displays:

- The path to the currently selected parameter file.
- Buttons for editing or replacing the configuration.
- Controls for starting the synthesis process.

Before running the simulation, users should verify that the displayed parameter file path is correct. If necessary, a different configuration file can be selected.

To start the simulation, click:

```
[Run Synthesis]
```

The program will then begin generating the synthetic population and calculating the expected contamination statistics.

4.4 Viewing the Results

The runtime depends on the selected model grid, survey area, and simulation settings. During execution, the program may require several minutes to complete.

Once the calculation is finished, the following button will become available:

```
[Show Image]
```

Selecting this option displays the output figure generated by the simulation. The figure includes the predicted contamination counts within the specified field of view and summarizes the expected population of contaminating objects based on the selected survey parameters.

Users should wait until the synthesis process has completed before attempting to view the results.

4.5 Program Workflow

The overall workflow of the automatic setup mode is summarized below:

```
   Do you already have a YAML parameter file?
                     |
       +-------------+--------------------+
       |                                  |
      Yes                                 No
       |                                  |
Select YAML File                  Create New YAML
       |                                  |
Load Configuration                        |
       |                                  |
Use this configuration directly?          |
       |                                  |
+------+------+                           |
|             |                           |
Yes           No                          |
|             |                           |
|      Open YAML Editor                   |
|             |                           |
|      Modify Parameters                  |
|             |                           |
+------+------+                           |
       |                                  |
Save Configuration --------------------- +
       |
Run Synthesis
       |
Calculation Complete
       |
Show Results
```

This workflow enables both first-time and experienced users to configure, execute, and analyze simulations without manually editing program files.

5 Simulation method and Output file

The output files are divided into three main stages:

1. Before the selection method,
2. During the selection method,
3. After the selection method.

5.1 Pre-selection Catalog Generation

Before applying any color-selection criteria, the program constructs a series of photometric catalogs that serve as the basis for all subsequent simulations. During this stage, two files are generated for each metallicity and model grid:

`mag_{filters_program}_m{mh_list}`

and

`flux_zp25_{filters_program}_m{mh_list}`

The first file contains AB magnitudes, while the second contains fluxes normalized to a zero point of 25. Each catalog includes the following parameters:

`name`, `logg`, `Teff`, `mh`, `cto`, `logkzz`, PH3, `Radius_pc`, `SpT`, and the corresponding fluxes or magnitudes in each selected filter.

Only the filters requested by the user are retained in the final output catalogs. However, the original model libraries contain synthetic photometry for all supported filters available in the program.

The currently supported filter sets are:

HST `F098M, F105W, F110W, F125W, F140W, F160W, F300X, F350LP, F435W, F475W, F475X, F555W, F600LP, F606W, F625W, F775W, F814W, F850LP`

JWST `F070W, F090W, F115W, F150W, F200W, F277W, F300M, F356W, F410M, F444W, F480M`

Roman `F062, F087, F106, F129, F146, F158, F184, F213`

5.1.1 Construction of the Parent Catalog

The parent catalog contains all ultracool dwarf (UCD) models together with their physical and photometric properties. The catalog is assembled from the selected atmospheric model grid (Sonora Bobcat or Sonora Elf Owl) and includes spectral type, effective temperature, surface gravity, metallicity, radius, and synthetic photometry.

Spectral types are assigned using the $T_{\mathrm{eff,SpT}}$ relation presented in (Kirkpatrick et al., 2021, Figure 22). Stellar radii are obtained by interpolating the Sonora Bobcat evolutionary models as a function of T_{eff} and $\log g$.

Since the Elf Owl atmosphere models do not provide independent evolutionary tracks, the Bobcat evolutionary tables are used to estimate radii. Furthermore, because the Bobcat evolutionary models are only available for metallicities in the range $[\mathrm{M/H}] \in [-0.5, +0.5]$, Elf Owl models outside this range are assigned radii interpolated from the nearest available Bobcat metallicity grid (-0.5 or $+0.5$).

Synthetic fluxes are computed by integrating the model spectra through the transmission profiles of the selected filters. Transmission curves for *HST*, *JWST*, and *Roman* instruments are included in the package and were obtained from the Spanish Virtual Observatory Filter Profile Service[3].

[3] `https://svo.cab.inta-csic.es`

5.1.2 Luminosity Statistics and Distance Limits

To estimate the number of detectable UCDs, the program computes luminosity statistics for each spectral type from late-M through Y dwarfs.

For a given reference filter x, the quantity $f_x r^2$

is calculated for every model, where f_x is the surface flux density in the reference filter and r is the stellar radius. For each spectral type, the median and 1σ dispersion of this quantity are then computed.

These luminosity statistics are used to determine the minimum and maximum distances over which objects of a given spectral type contribute to a particular apparent-magnitude bin. The expected number of objects is calculated from

$$N(\mathrm{SpT}) = \Delta\Omega \int_{d_{\min}(\mathrm{SpT})}^{d_{\max}(\mathrm{SpT})} \rho(r), r^2, dr, \tag{1}$$

where $\Delta\Omega$ is the survey solid angle and $\rho(r)$ is the spatial density distribution.

To ensure completeness, relatively broad distance intervals are adopted. The procedure begins by estimating the intrinsic distance spread for each spectral type from the Elf Owl model distribution. The theoretical flux corresponding to the magnitude-bin boundaries, $m_{\mathrm{bin}} \pm 0.5\Delta m$

is then calculated, where Δm is the bin width.

The minimum distance is defined using the lower 1σ distance bound and the maximum theoretical luminosity, while the maximum distance is determined using the upper 1σ distance bound and the minimum theoretical luminosity. This approach allows the simulation to sample a sufficiently broad volume for each spectral type and reduces edge effects at bin boundaries.

5.1.3 Generation of Simulated Observations

The final stage generates synthetic observations for each magnitude bin. Output files are stored as

```
mag_{corresponding_to_magnitude_shift}
```

For each magnitude bin, every UCD model is shifted so that its apparent magnitude in the reference filter matches the center of the bin. Fluxes in all other filters are scaled consistently using the same distance correction. flux density per unit frequency f_ν in cgs units (erg s^{-1} cm^{-2} Hz^{-1}) as

$$\text{Poisson Noise}_e = \sqrt{\frac{c \cdot f_\nu \cdot \mathtt{EXPTIME}}{\mathtt{PHOTPLAM}^2 \cdot \mathtt{PHOTFLAM}} + \left(\frac{c \cdot f_{1\sigma,\nu} \cdot \mathtt{EXPTIME}}{\mathtt{PHOTPLAM}^2 \cdot \mathtt{PHOTFLAM}}\right)^2}. \tag{2}$$

`PHOTPLAM`, `PHOTFLAM` and `EXPTIME` are common parameters in the header of HST images. Specifically, `PHOTFLAM` is the flux (f_λ) that would produce 1 electron per second, with a unit of (erg cm^{-2} s^{-1} Å^{-1})$(e/s)^{-1}$. `PHOTPLAM` is the pivot wavelength in angstrom. `EXPTIME` is the total exposure time is in seconds. Finally, the speed of light c is expressed in Å s^{-1}.

Finally, the simulated observed fluxes are generated by perturbing the noiseless fluxes with Gaussian random noise using the calculated uncertainty in each filter. These catalogs constitute the input data used by the subsequent color-selection and contamination-analysis stages of the pipeline.

5.2 Contamination Selection Method

After the simulated catalogs have been generated, the program evaluates the contamination rate of ultracool dwarfs (UCDs) using two complementary approaches:

1. **High-redshift galaxy selection**
2. **UCD number-density calculation within the Milky Way**

The results from these analyses are combined to estimate the expected contamination from UCDs in high-redshift galaxy surveys.

5.2.1 High-redshift Galaxy Selection

This stage identifies simulated UCDs that would be misclassified as high-redshift galaxy candidates. Two independent selection techniques are implemented:

- Photometric redshift (*photo-z*) selection
- Near-infrared (NIR) color selection

The output files generated by these methods are stored in the corresponding subdirectories:

```
Eazy/
```

and

```
Color/
```

within each magnitude-shift directory.

Photometric Redshift Selection The photometric-redshift analysis is performed using the EAzY package.

For each simulated catalog, EAzY is executed twice:

1. A galaxy-template fit is performed to estimate the best-fitting redshift of each object.
2. A second fit compares galaxy and stellar templates using their respective χ^2 values to determine whether the object is more likely to be classified as a galaxy or a star.

It is important to note that all input sources are simulated UCDs. Therefore, any object classified as a galaxy is considered a contaminant.

The `Eazy` directory contains:

- The input catalog used by EAzY.
- Standard EAzY output products (e.g., `zout.fits`).
- A summary table containing:
  - Object identifiers.
  - Best-fit galaxy redshift (z_{best}).
  - χ^2 values from galaxy-template fitting.
  - χ^2 values from stellar-template fitting.
  - Redshift posterior distributions, $P(z)$.

These outputs allow users to investigate why specific UCDs are identified as potential high-redshift galaxies.

5.2.2 Color Selection

The NIR color-selection module applies published high-redshift galaxy selection criteria from:

- Bouwens et al. (2015)
- Roberts-Borsani et al. (2022)

For each simulated object, the program evaluates whether it satisfies the color cuts associated with each redshift-selection criterion.

The resulting output table contains the object identifier together with a Boolean flag for every implemented color-selection criterion. An example structure is:

```
ID   z8_098   z8_105   z9_105   ...
```

where each column corresponds to a particular dropout-selection criterion.

The values stored in these columns are:

- `True`: the object satisfies the selection criterion.
- `False`: the object does not satisfy the selection criterion.

These tables allow users to identify which UCDs would contaminate specific high-redshift galaxy samples.

5.2.3 Number Density of UCDs in the Milky Way

In addition to the selection analysis, the program computes the expected number density of UCDs within the Milky Way using the Galactic structure model of Aganze et al. (2022). The calculation is based on Equation 1.

Results are saved in:

```
Analysis/Number_density/
```

For each magnitude-shift catalog, the program generates a summary file named:

`summary_mag_{magshift}.tsv`

Each file contains the estimated density and object counts for every spectral type and Galactic component. Typical columns include:

`spectral_type, d_min, d_med, d_max, curtemp, rhor2_thindisk, rhor2_thickdisk, rhor2_halo, N_thindisk, N_thickdisk, N_halo, N_browndwarf_total`

along with their associated uncertainties.
The uncertainty columns follow the naming convention:

- `eneg_*`: lower (negative) uncertainty.
- `epos_*`: upper (positive) uncertainty.

The resulting tables provide the expected number of UCDs in the:

- Thin disk,
- Thick disk,
- Halo,

as well as the total predicted number of contaminating brown dwarfs for each spectral type and magnitude interval.

The contamination-selection stage combines the classification results from the photometric-redshift and color-selection analyses with the Galactic number-density calculations. Together, these outputs provide an estimate of the expected contamination from ultracool dwarfs in high-redshift galaxy surveys and identify which spectral types contribute most significantly to the contamination rate.

5.3 Post-selection Analysis

After the contamination-selection stage is completed, the program proceeds to the final analysis stage. This stage combines the outputs from the photometric-redshift selection, color-selection criteria, and Galactic number-density calculations to estimate the expected contamination rates as a function of apparent magnitude.

The program generates a summary table containing quantities such as:

`mag, Number_density, N_err_lt, N_err_gt, fraction_eazy, err_eazy, color_z8_105, err_color_z8`$_1 05, both\_z8\_105, err\_both\_z8\_105, \ldots$

where:

- `mag` is the center of the apparent-magnitude bin.
- `Number_density` is the predicted number of UCD contaminants within that magnitude bin.

- `N_err_lt` and `N_err_gt` are the lower and upper uncertainties on the predicted number density.
- `fraction_eazy` is the fraction of simulated UCDs classified as high-redshift galaxies by the EAzY photometric-redshift selection.
- `err_eazy` is the uncertainty associated with this fraction.
- `color_criteria` is the fraction of UCDs that satisfy a specific NIR color-selection criterion.
- `err_color_{criteria}` is the corresponding uncertainty.
- `both_{criteria}` is the fraction of UCDs that simultaneously satisfy both the EAzY selection and the specified color-selection criterion.
- `err_both_{criteria}` is the uncertainty on this combined fraction.

The contamination fraction is calculated independently for each magnitude bin and selection method. This allows users to determine how the expected contamination varies with survey depth and selection strategy.

In addition to the summary tables, the program automatically generates diagnostic figures and contamination-rate plots. These visualizations summarize the expected number of contaminating UCDs as a function of magnitude and selection criteria.

Once the analysis is complete, the figures can be displayed directly through the graphical user interface by selecting:

```
[Show Image]
```

The generated plots provide a visual summary of the contamination predictions and can be used to compare different selection methods, survey configurations, and filter combinations.

Figure 1 shows an example contamination analysis produced by the ENZO program for a representative survey field.

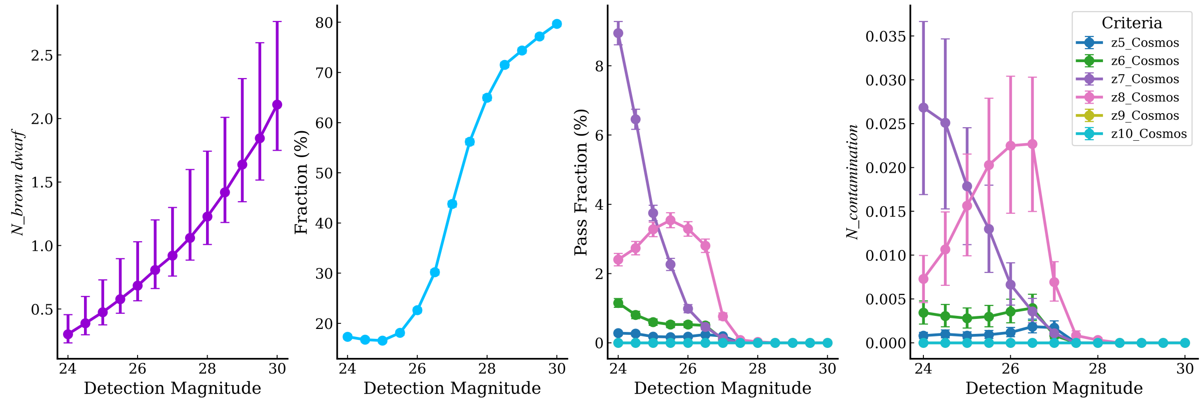

Figure 1: Example output from ENZO using the fiducial model setup for the COSMOS-1000+0224 field with the $z \sim 5-8, 10$ galaxy selection criteria from Bouwens et al. (2015) and ~ 9 adopted by Roberts-Borsani et al. (2022). The first panel shows the total number density of UCDs of all spectral types as a function of F160W magnitude, which depends on the sky position of the field. The second panel shows the fraction of UCDs for which the galaxy template is preferred over the stellar template based on a χ^2 comparison. The third panel shows the fraction of UCDs that satisfy the remaining $z \sim 5 - 10$ galaxy selection criteria, including the NIR color–color cuts, signal-to-noise ratio requirements in different bands, and photometric redshift criteria. The final panel shows the expected number of contaminants in $z \sim 5 - 10$ sample in the field as a function of magnitude.

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